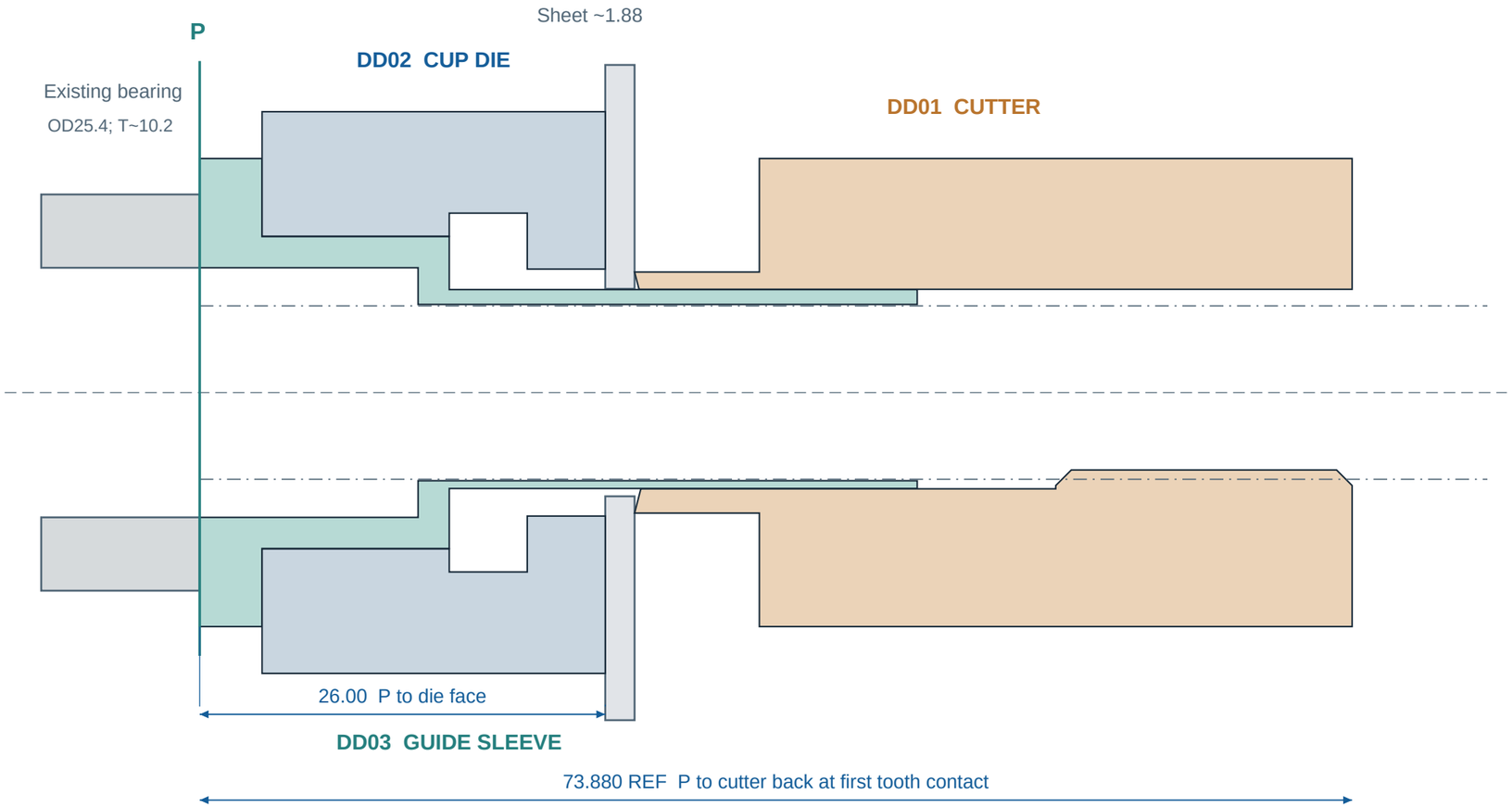
DOUBLE-D PUNCH / ASSEMBLY

Three custom parts; reuse the complete 7/16-20 stud and flat-faced bearing. Section is through the key (+Y).



CONTROLLED ASSEMBLY LIMITS

Thread	7/16-20 UNF; existing stud
Full overlap	17.00 min in the modeled stroke
Pilot hole	D13.30 +/-0.05; deburr first
Hole target	D15.4-15.5; AF13.6-13.7
Cutting clearance	0.19 normal per side, nominal
Maximum entry	5.00 tooth-tip entry into die
Maximum stroke	6.880 from first tooth contact
At maximum entry	P to cutter back = 67.00 REF
Sleeve clearance	2.00 to cutter bore step at end

PARTS / ONE TOOL ASSEMBLY

DD01 / 1	One-piece cutter; S7, 54-56 HRC
DD02 / 1	One-piece cup die; A2, 58-60 HRC
DD03 / 1	One-piece indexed guide; S7, 52-54 HRC
Existing / 1 each	Your draw stud and its complete bearing stack; neither is modified.

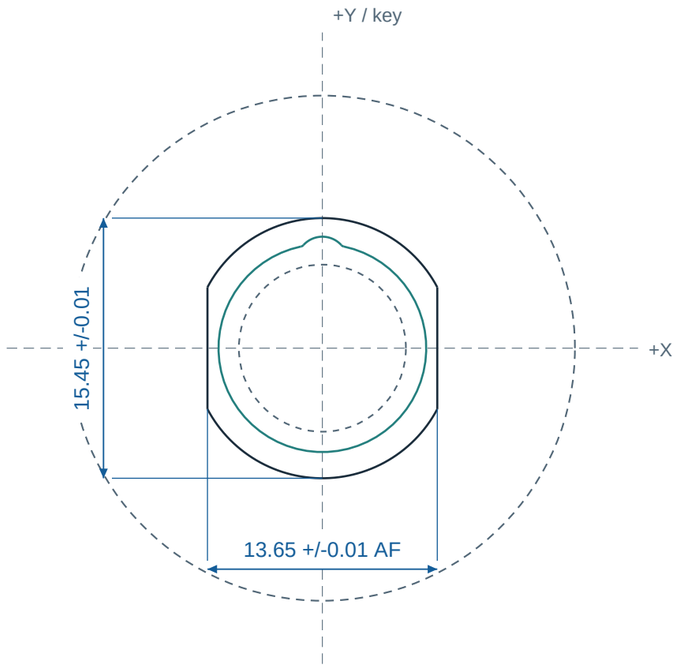
All custom-part profiles share the same axis. Key bump points +Y; finished-hole flats are parallel to Y.

Hold cup AF30 against rotation; turn the existing stud. The guide clocks the cutter to the cup.

Stock bearing bore, stud head and shoulder are not reverse engineered. Gray stock shapes are reference envelopes.

DD01 / ONE-PIECE THREADED CUTTER

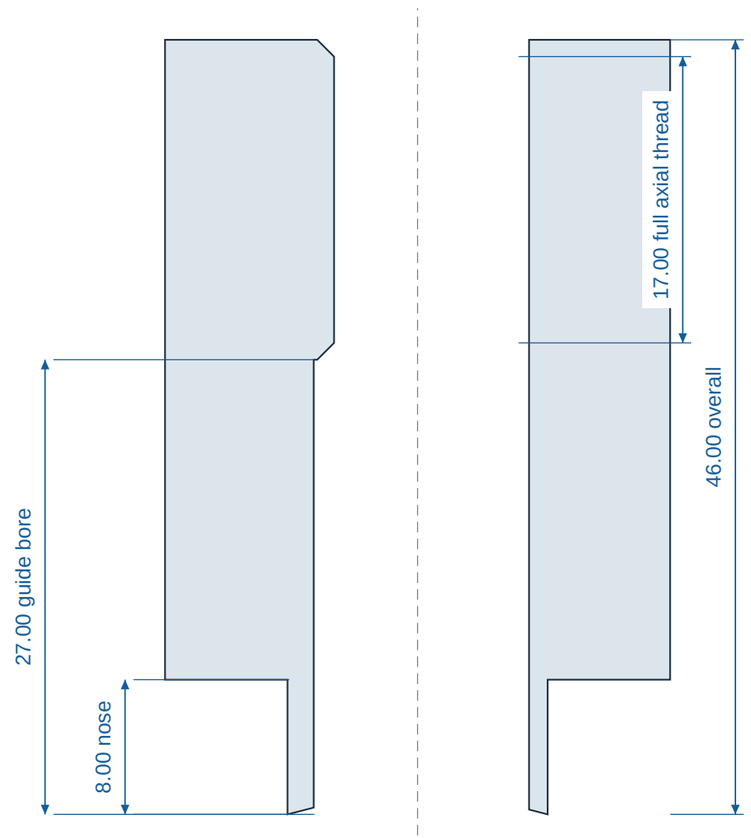
S7 tool steel; harden and temper to 54-56 HRC. Critical profiles finished after heat treatment.



CUTTING PROFILE / XY

Dashed outer circle: D30 body. Green: guide socket.

Through keyway continues into the rear thread.



SECTION YZ / X=0

BALANCED TWO-TOOTH FACE

Z=2.00 x (1 - |Y|/7.725) on the nose face. Highest tips at Y=+/-7.725, Z0; valley at Y0, Z2.00. Two planar grinds, 14.51 deg REF.

Overall / body	Z0-46.00; body D30, Z8-46
Cutting outline	D15.45 / AF13.65; DD01_Cutting_outline.dxf
Guide socket	DD01_Guide_socket_profile.dxf, Z0-27.00; nominal outward normal offset 0.02 from DD03 small stem
Through keyway	Through Z0-46. Union of slot X+/-2.10, Y0-5.80 with socket region Y>=5.80; see keyway DXF
Core bore	D9.92 tap-drill representation, Z27-46
Thread	7/16-20 UNF-2B; interrupted by keyway
Thread leads	2 x 45 deg cones: D11.92 mouth to D9.92 core, 1.00 axial
Full thread	Z28.00-45.00; GO/NO-GO check after hardening
Wrench flats	AF24; planes X=+/-12; Z36-46
Edge finish	Cutting edges sharp, R0.02 max; other edges break 0.10 max

Z coordinates originate at the leading tooth tips, not the relieved bore mouth. All profiles centered on the thread axis.

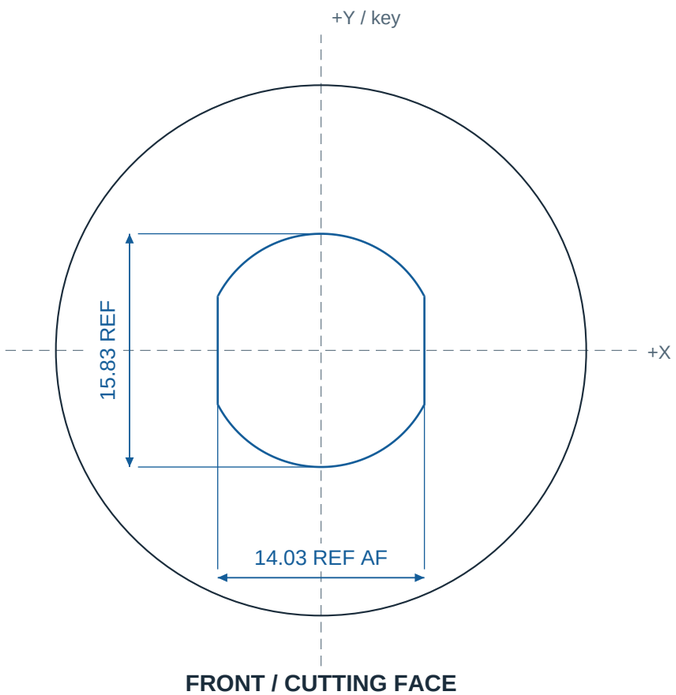
Flats of the DD outline are X=+/-6.825. Key bump is +Y. Keep their clocking within 0.10 deg.

STEP thread is simplified. The 7/16-20 UNF-2B callout controls machining; the STEP bore is not a clearance hole.

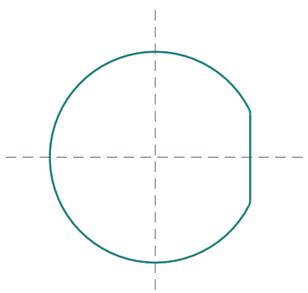
Profile error +/-0.01 normal; guide/socket matched clearance 0.01-0.03 normal. Concentricity of socket and thread axes 0.02 TIR max. Straight nose has no back taper in this low-volume prototype.

DD02 / ONE-PIECE CUP DIE

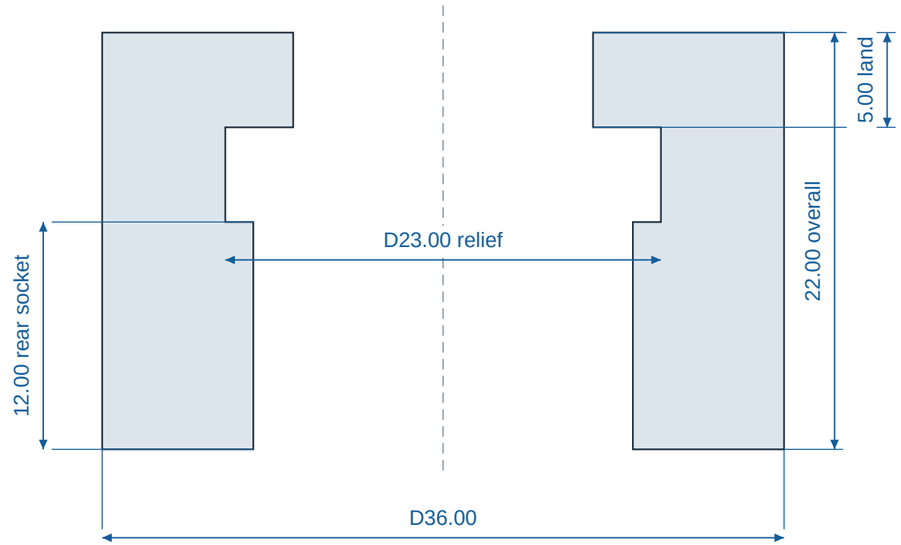
A2 tool steel; harden and temper to 58-60 HRC. Rear socket indexes on guide flange/register.



FRONT / CUTTING FACE



REAR SOCKET / +X FLAT



SECTION YZ / X=0

The YZ section passes through X0 and does not show the +X rear indexing flat. The rear-view DXF controls that flat and its R1.02 nominal corner blends.

Outside / length	D36.00 x 22.00
Rear socket	DD02_Rear_socket_profile.dxf; Z0-12.00
Rear nominal form	Normal offset 0.02 from DD03 rear register; bounding width 19.04 REF
Internal relief	D23.00; Z12.00-17.00
Cutting land	DD02_Die_opening.dxf; Z17.00-22.00
Die opening	Exact outward normal offset 0.19 from cutter; D15.83 / AF14.03 REF
Wrench flats	AF30; planes X=+/-15; Z0-12
Clocking	Rear D-flat is +X; cutting flats parallel to Y
Face / axis	Front and rear faces parallel 0.03; cutting profile axis to rear socket axis 0.02 TIR
Edge finish	Cutting edge sharp, R0.02 max; no front chamfer

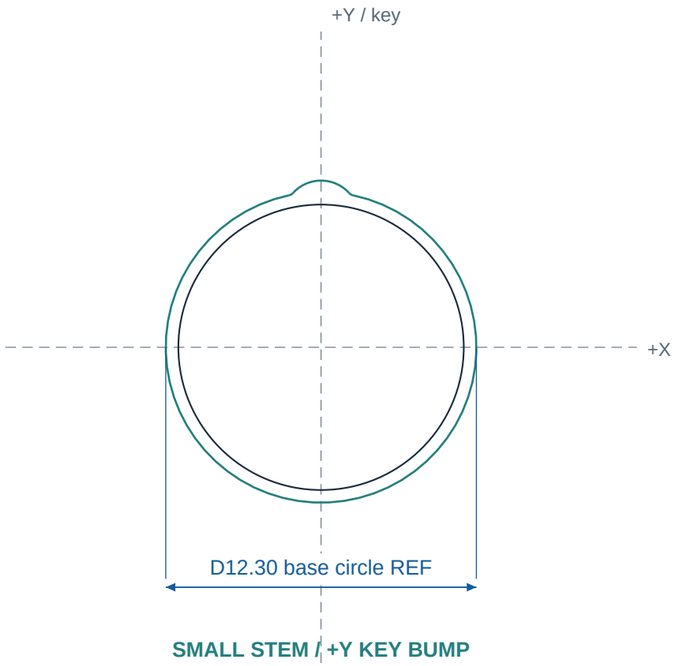
Cup rear Z0 seats on the guide flange at assembly Z4. Cup front is assembly Z26.

The four arc-to-flat transitions are offset arcs, not intersections made by simply enlarging a circle and rectangle. DXF is the controlling profile.

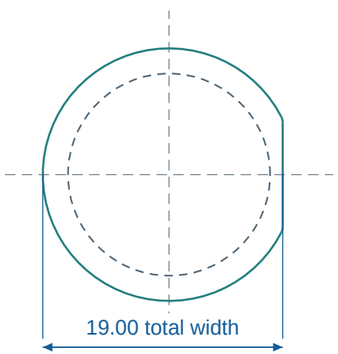
Use profile +/-0.01 normal. Match rear socket to actual hardened guide at 0.01-0.03 normal clearance.

DD03 / INDEXED GUIDE SLEEVE

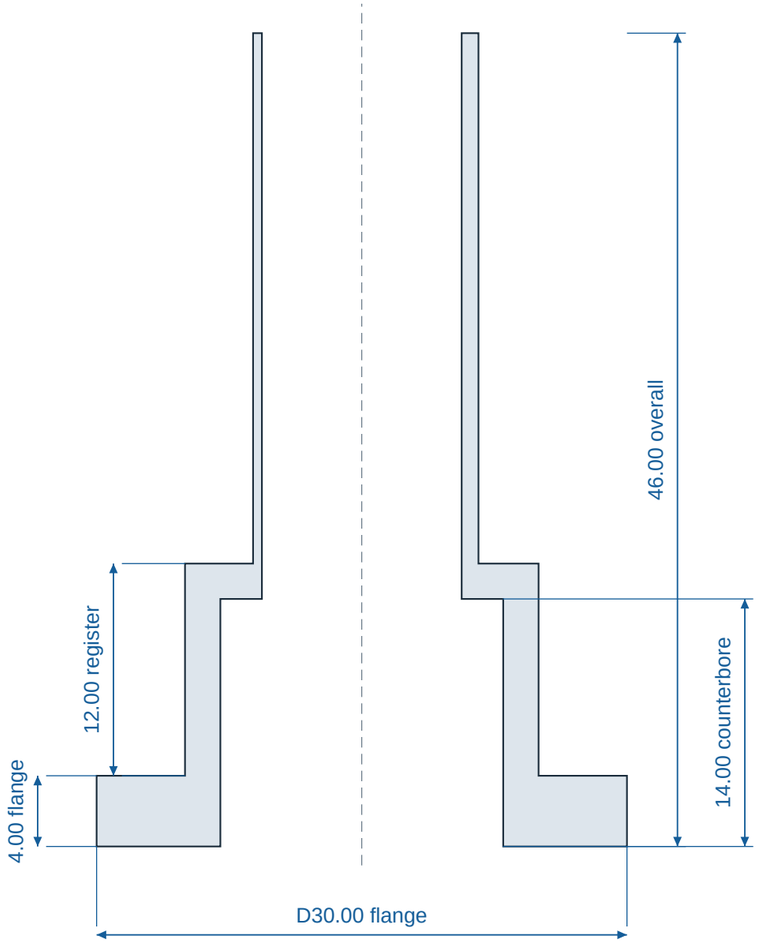
S7 tool steel; harden and temper to 52-54 HRC. This part carries clocking torque; it is not a draw stud.



SMALL STEM / +Y KEY BUMP



LARGE REGISTER / FLAT X=+9



SECTION YZ / X=0

Flange	D30.00; Z0-4.00; Z0 = bearing face P
Large register	D20 base circle clipped at X=+9.00; R1.00 blends; Z4-16
Small stem	DD03_Small_guide_profile.dxf; Z16-46
Stem construction	Union D12.30 at origin with R1.50 bump centered X0 Y+5.10; R0.30 root blends
Clearance envelope	D13.20 enclosing circle REF
Central bores	D16.00 +0.10/0, Z0-14; D11.30 +0.05/0, Z14-46
Profile / fit	Guide profile +/-0.01 normal; match both sockets at 0.01-0.03 normal clearance
Clocking	Large register flat +X; small key bump +Y; relative angle 90.00 +/-0.10 deg
Faces / bores	Bearing seat flat 0.03; bore/profile axes coaxial 0.02 TIR
Entry edges	Break exposed guide tip/bore edges 0.10 max; keep full profile below edge break

Minimum nominal small-stem wall is 0.50 mm. No screw or clamp bears on this stem.

Verify your stud shoulder passes the D16 counterbore and ends before Z14. The measured first full thread starts at Z12.70.

Reusing the complete stock bearing avoids needing its unmeasured internal bore dimensions.

FABRICATION / FIT / FIRST COUPON

Complete geometry is specified; load qualification and hole-process validation remain physical checks.

MANUFACTURING ORDER

1. Make the three blanks in the specified annealed tool steels. Rough all geometry; provide finish allowance suited to the heat-treatment process.

2. DD01: machine the D30 body and nose, drill/tap the rear thread, bore the front guide socket, cut the through keyway, and make the two planar shear grinds. Keep the interrupted thread free of wire-EDM recast/burrs.

3. DD02: bore the rear pocket, make the D23 internal relief, and leave a 5 mm front land. Machine the rear indexing socket. Wire EDM the final DD die opening through the front land; its wire path clears the larger rear opening.

4. DD03: machine flange, large rear register and keyed small stem. Bore the D16 shoulder clearance and D11.30 thread clearance. Finish guide roots with the specified R0.30 blends.

5. Harden/temper through a controlled tool-steel process. Protect against decarburization. Finish profiles, faces and matched fits after hardening. Check hardness and the thread with GO/NO-GO gauges.

The functional DXFs are XY profiles in millimeters. The PDF axial Z stations establish each feature depth. STEP helices are omitted: thread callout, lead geometry and keyway must all be respected.

CALCULATED GEOMETRY CHECKS

Round-trip STEP	Each custom part reimported as one valid solid
Aligned travel	No solid interference at eight sampled positions through the 6.880 mm stroke
Thread overlap	17.00 mm full axial window remains on measured full stud threads
Guide end gap	2.00 mm to the guide-bore step at maximum specified entry
Flat-face force reference	31.07 kN / 3.49 US tons
Minimum nose net area	58.34 mm2 from actual CAD profile
Mean compression reference	533 MPa at the force reference

WHAT THESE CHECKS DO NOT ESTABLISH

The force reference uses perimeter x 1.8796 mm x 344.7 MPa assumed steel shear strength. The shear grind may reduce peak load; no reduction is credited. This is not a rated load or a tightening-torque specification.

Tooth bending, key contact, heat-treatment defects, stud proof load and thread stripping are not verified by a collision check or mean-stress calculation. The thin nose and guide stem make these real qualification items.

Before manufacturing, verify the stock shoulder clears D16 over the first 14 mm from P. Before enclosure use, confirm free dry travel, correct clocking and a coupon hole within D15.4-15.5 / AF13.6-13.7 after light burr removal.

If the coupon is undersize/oversize, tears at the flats, or shows excessive burr, revise the punch target and/or die clearance. Do not change the connector-required finished-hole limits.

REFERENCE BASIS

TE T4171310004-001 customer drawing B2: required panel cutout. Owner measurements control the reused stud/bearing interface.

Mate LIT00682 Rev E: piercing clearance and perimeter/thickness/shear-strength force method. Clearance 0.19 per side is an initial tool-design choice.

Carpenter S7 / Hudson A2 technical data: material selection and controlled heat treatment. Heat treater establishes the alloy-specific cycle.

ASSEMBLY / FIRST COUPON RECORD

Use the actual enclosure material; accept the hole against the connector drawing, not against nominal tool sizes.

ASSEMBLY AND USE

1. Drill a D13.30 +/-0.05 pilot hole in a coupon of the actual enclosure steel. Deburr both faces. Confirm the finished pilot remains below the 13.6 mm minimum finished flat spacing.

2. Keep the existing bearing on the stud. Add the guide flange against flat face P, then seat the cup on the large D register. Insert the small keyed stem through the pilot hole.

3. Hand-engage the stud in the cutter. Align its key socket as the sleeve enters; turn the stud to advance the cutter. Engage the guide before either tooth touches the sheet.

4. Hold the cup AF30 and turn the stud with a hand ratchet. Stop once the hole is complete. Do not exceed 5 mm tooth entry; the drawing gives the corresponding P-to-back distance. No impact-wrench qualification is provided.

5. Back the stud out, remove the cutter and sleeve, and release the slug. Inspect the cutting edges, key, thread and finished hole before further use.

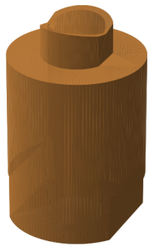
RECORD BEFORE ENCLOSURE USE

Actual sheet thickness	_____ mm
Stock shoulder OD	_____ mm; must clear D16
Guide dry fit / clocking	_____; free through intended stroke
Stud / bearing condition	_____
Coupon major diameter	_____ mm; accept 15.4-15.5
Coupon across flats	_____ mm; accept 13.6-13.7
Burr / tearing / slug release	_____
Cutting edges / key / thread	_____ after coupon
Further action	_____

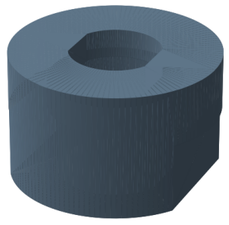
Inspect the complete hole profile, including the four arc-to-flat transitions. A caliper reading at one height does not establish the whole punched edge; check connector fit and antirotation engagement.

Double-D punch: three custom parts / existing stud and bearing reused

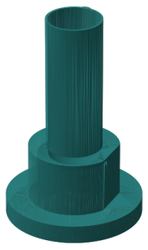
DD01 Cutter



DD02 Cup die



DD03 Guide sleeve



REV A | DIMENSIONED PROTOTYPE | Threads simplified in STEP; machining callout controls.